

AUCA

Statistical Learning, Regression and Classification

DATA MINING WORK

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Introduction

These solutions follow James, Witten, Hastie, and Tibshirani, *An Introduction to Statistical Learning with Applications in R*, 2nd edition (corrected June 2023). Conceptual answers use the definitions of regression, classification, inference, and prediction from Chapter 2. Applied answers use the `Auto` data and linear-model functions in R. All numerical results reported in this submission were produced with R 4.4 and ISLR2.

1 Chapter 2: Statistical Learning

1.1 Exercise 2: Classification versus Regression; Inference versus Prediction

For each scenario, the problem is classified according to the response variable, the main objective, and the numbers of observations and predictors.

(a) CEO salary of the top 500 U.S. firms

This is a **regression** problem because CEO salary is a continuous quantitative response. The main goal is **inference**, because the interest is in understanding which factors affect CEO salary rather than simply forecasting a new CEO's salary. The sample contains $n = 500$ observations, with $p = 3$ predictors: profit, number of employees, and industry. CEO salary is the response and is therefore not counted among the predictors.

(b) Success or failure of a new product

This is a **classification** problem because the response is binary: success or failure. The main goal is **prediction**, since the practical objective is to predict whether a product that has not yet been launched will succeed. Here $n = 20$ previously launched products are available and there are $p = 13$ predictors, including price charged, marketing budget, competition price, and ten other variables. Since $n = 20$ and $p = 13$, this is a small-sample, relatively high-dimensional problem.

(c) Weekly percentage change in the USD/Euro exchange rate

This is a **regression** problem because the response is a continuous percentage change. The main goal is **prediction**: the objective is to forecast the exchange-rate change using

weekly changes in stock markets. There are $n = 52$ weekly observations for 2012 and $p = 3$ predictors: the percentage changes in the U.S., British, and German markets.

Table 1: Summary of Exercise 2

Scenario	Problem type	Goal	n	p
CEO salary	Regression	Inference	500	3
Product launch	Classification	Prediction	20	13
USD/Euro change	Regression	Prediction	52	3

1.2 Exercise 9: The Auto Data Set

The `Auto` data set in `ISLR2` already has missing horsepower values removed. Five rows were omitted, leaving $n = 392$ complete observations and nine variables. The analysis below therefore uses the complete data frame without additional deletion.

```

1 library(ISLR2)
2 data(Auto)
3 dim(Auto) # 392 9
4 anyNA(Auto) # FALSE

```

(a) Quantitative versus qualitative predictors

The variables are `mpg`, `cylinders`, `displacement`, `horsepower`, `weight`, `acceleration`, `year`, `origin`, and `name`. The quantitative variables are `mpg`, `cylinders`, `displacement`, `horsepower`, `weight`, `acceleration`, and `year`. The qualitative variables are `origin` and `name`. `Origin` is coded as American, European, and Japanese; the numerical codes do not represent a continuous scale. `Name` is a factor containing manufacturer/model labels.

Table 2: Variable types in the Auto data

Variable	Storage/description	Role
mpg	numeric	Quantitative (response in part f)
cylinders	integer	Quantitative; few levels
displacement	numeric	Quantitative
horsepower	integer	Quantitative
weight	integer	Quantitative
acceleration	numeric	Quantitative
year	integer	Quantitative; few levels
origin	integer codes	Qualitative
name	factor	Qualitative

Cylinders and year are borderline cases because they have relatively few distinct values. For the range, mean, and standard-deviation calculations below, they are treated as quantitative. Origin should not be treated as a continuous predictor.

(b) Range of each quantitative predictor

The range is calculated as $\max(x) - \min(x)$.

Table 3: Range of quantitative variables in the full Auto data ($n = 392$)

Predictor	Minimum	Maximum	Range
mpg	9.0	46.6	37.6
cylinders	3.0	8.0	5.0
displacement	68.0	455.0	387.0
horsepower	46.0	230.0	184.0
weight	1613.0	5140.0	3527.0
acceleration	8.0	24.8	16.8
year	70.0	82.0	12.0

MPG ranges from 9.0 to 46.6 miles per gallon. Engine size and vehicle mass show much larger numerical ranges, especially displacement and weight.

(c) Mean and standard deviation of each quantitative predictorTable 4: Mean and standard deviation of quantitative variables ($n = 392$)

Predictor	Mean	SD
mpg	23.45	7.81
cylinders	5.47	1.71
displacement	194.41	104.64
horsepower	104.47	38.49
weight	2977.58	849.40
acceleration	15.54	2.76
year	75.98	3.68

A typical car in the sample obtains about 23.45 mpg, has approximately 194 cubic inches of displacement, 104 horsepower, and weighs about 2,978 pounds. The relatively large standard deviations of displacement, horsepower, and weight indicate substantial variation between compact cars and heavier, more powerful vehicles.

(d) Range, mean, and SD after dropping observations 10 through 85

Rows 10 through 85 contain 76 observations. Removing these rows leaves $392 - 76 = 316$ observations.

```

1 Auto_sub <- Auto[-c(10:85), ]
2 nrow(Auto_sub) # 316
3 sapply(Auto_sub[quant], range)
4 sapply(Auto_sub[quant], mean)
5 sapply(Auto_sub[quant], sd)

```

Table 5: Summary after removing observations 10–85 ($n = 316$)

Predictor	Min	Max	Range	Mean	SD
mpg	11.0	46.6	35.6	24.40	7.87
cylinders	3.0	8.0	5.0	5.37	1.65
displacement	68.0	455.0	387.0	187.24	99.68
horsepower	46.0	230.0	184.0	100.72	35.71
weight	1649.0	4997.0	3348.0	2935.97	811.30
acceleration	8.5	24.8	16.3	15.73	2.69
year	70.0	82.0	12.0	77.15	3.11

After deletion, mean mpg increases from 23.45 to 24.40, while mean horsepower decreases from 104.47 to 100.72 and mean weight decreases from 2977.58 to 2935.97. Mean year increases from 75.98 to 77.15. These changes indicate that the removed observations included many older and heavier vehicles. The descriptive statistics are therefore somewhat sensitive to the particular observations retained.

(e) Graphical investigation of the predictors

A combination of a scatterplot matrix, mpg-versus-predictor plots with a LOESS smoother, boxplots by origin and cylinders, and a Pearson correlation heatmap provides a broad view of the relationships.

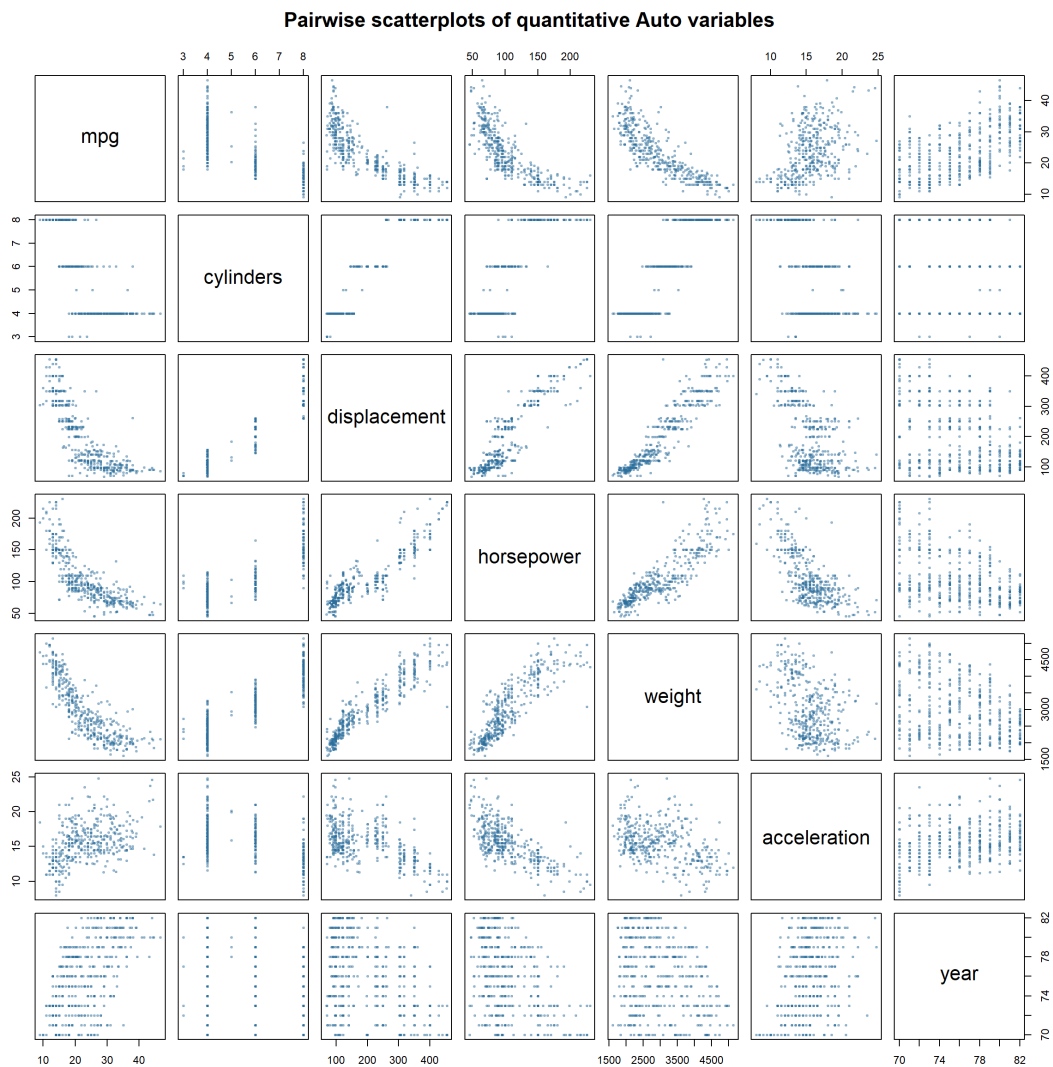


Figure 1: Scatterplot matrix of the quantitative Auto variables.

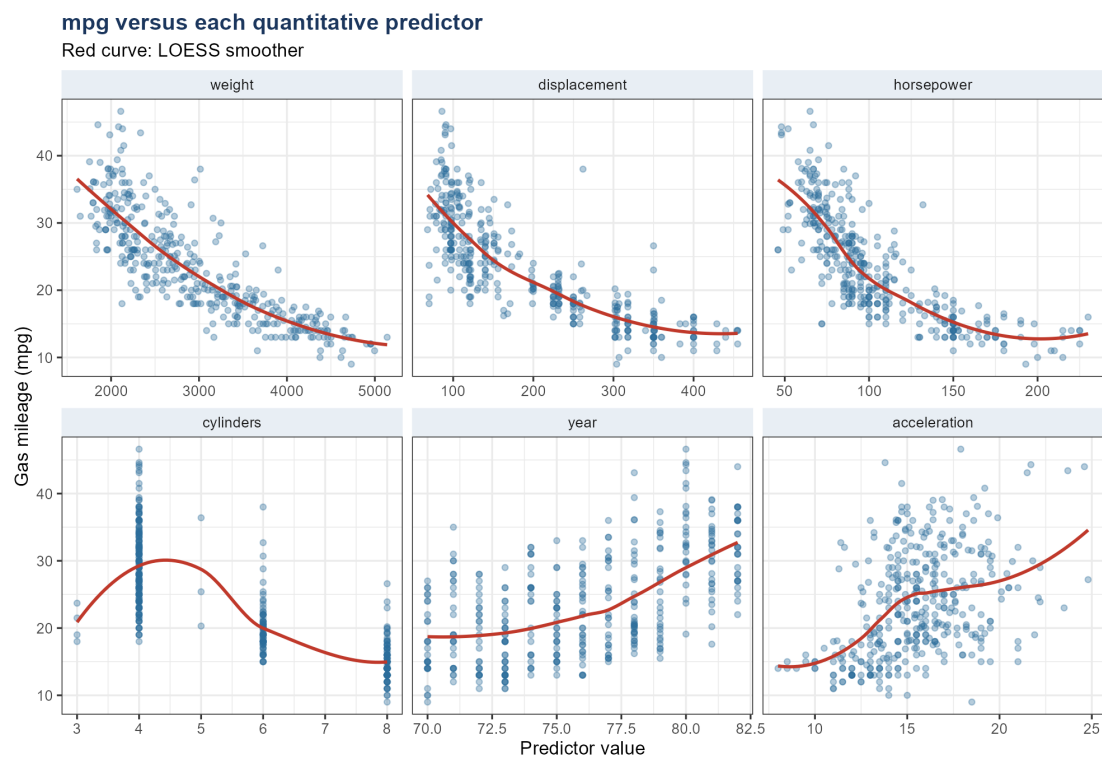


Figure 2: MPG against each quantitative predictor with a LOESS smoother.

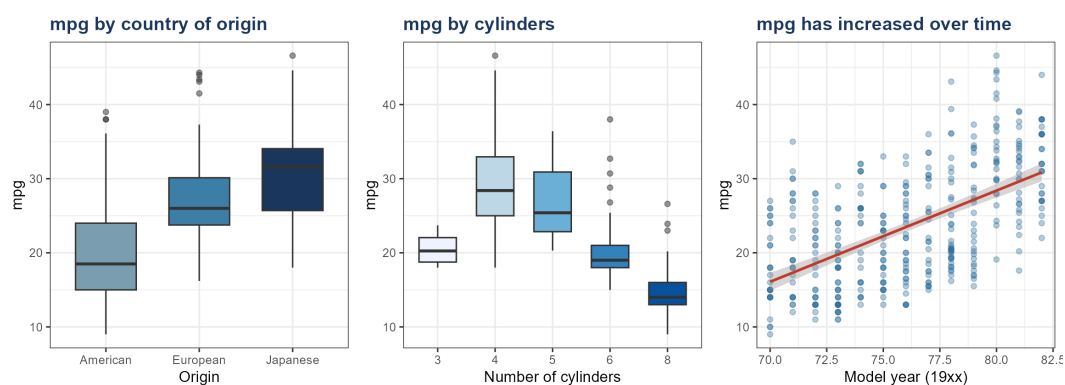


Figure 3: MPG by origin and cylinders, together with the upward trend with model year.

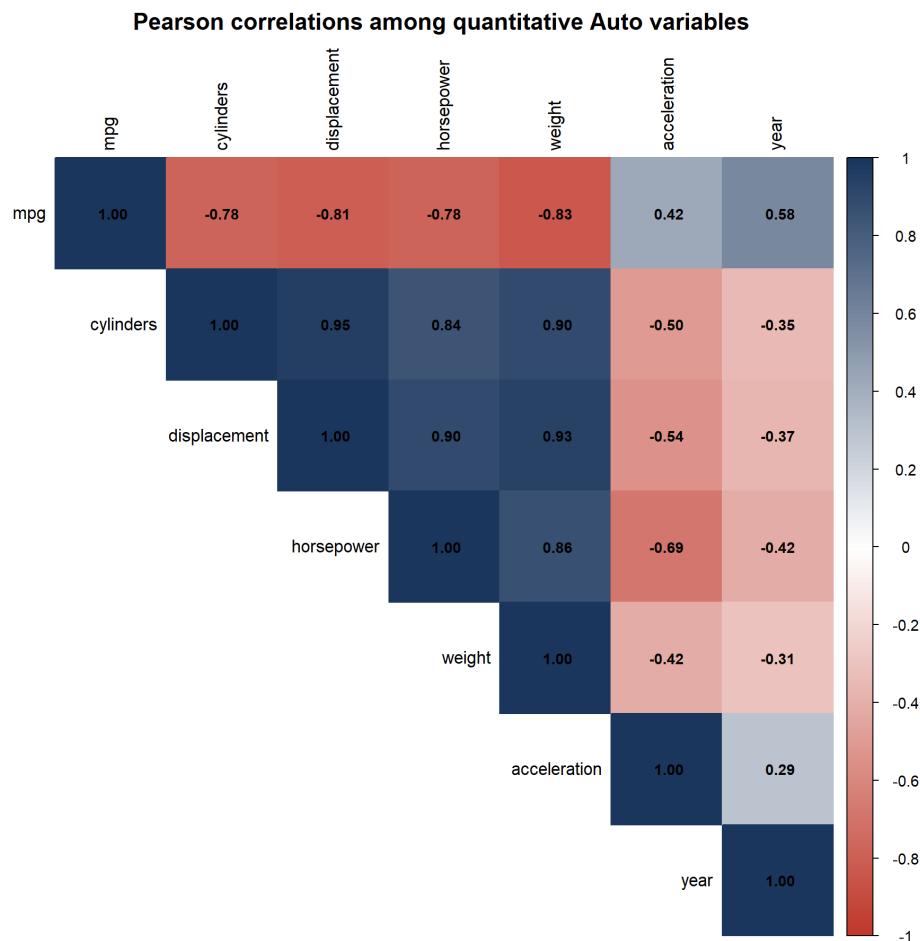


Figure 4: Pearson correlation matrix for quantitative variables.

The graphical analysis gives the following findings:

1. **Size and power are strongly related.** Cylinders, displacement, horsepower, and weight are strongly positively correlated, with pairwise correlations approximately between 0.84 and 0.95. These variables largely represent the same underlying “big car” factor.
2. **MPG decreases as cars become larger and more powerful.** Weight has the strongest linear association with mpg ($r = -0.832$), followed by displacement ($r = -0.805$), horsepower ($r = -0.778$), and cylinders ($r = -0.778$).
3. **Acceleration has a weaker relationship with mpg.** The correlation is approximately $r = 0.423$, so the relationship is positive but relatively noisy.
4. **Later model years are associated with better mileage.** The correlation between year and mpg is $r = 0.581$, showing an upward trend from 1970 to 1982.
5. **Origin matters.** American cars have the lowest mean mpg (20.03), European cars have a higher mean (27.60), and Japanese cars have the highest mean (30.45).

6. **Cylinders show a step-like pattern.** Four-cylinder vehicles dominate the high-mpg region, whereas eight-cylinder vehicles are concentrated at low mpg. Three- and five-cylinder observations are rare.

Table 6: MPG by origin

Origin	n	Mean mpg	SD mpg
American	245	20.03	6.44
European	68	27.60	6.58
Japanese	79	30.45	6.09

Table 7: MPG by number of cylinders

Cylinders	n	Mean mpg	SD mpg
3	4	20.55	2.56
4	199	29.28	5.67
5	3	27.37	8.23
6	83	19.97	3.83
8	103	14.96	2.84

(f) Variables that look useful for predicting mpg

Several predictors show clear associations with mpg and would be useful in a regression or nonparametric prediction model. The most informative quantitative variables are weight, displacement, horsepower, and cylinders, all of which have absolute correlations with mpg of approximately 0.78 or greater. However, these predictors are highly collinear, so a multiple regression containing all of them may produce unstable individual coefficient estimates even when overall model fit is good.

Year is also useful because its correlation with mpg is 0.581. Origin is useful as a categorical factor because Japanese and European vehicles tend to have higher mpg than American vehicles. Acceleration is weaker and partly acts as a proxy for horsepower. Name is not a useful regression predictor because it behaves almost like an identifier with many levels.

The plots also suggest mild nonlinearity, particularly at the high-mpg end. A linear model using weight or horsepower is a reasonable starting point, while a transformation such as $\log(\text{weight})$ or a spline could potentially improve prediction.

2 Chapter 3: Linear Regression

2.1 Exercise 2: KNN Classifier versus KNN Regression

K-nearest neighbors (KNN) uses the same neighborhood principle for both classification and regression. Given a query point x_0 , the K training observations closest to x_0 are identified, normally using Euclidean distance after appropriate scaling. The difference is in the type of response and how the neighbors are summarized.

KNN classifier

For qualitative response Y , the method estimates the proportion of each class among the K nearest neighbors:

$$\hat{P}(Y = j \mid X = x_0) = \frac{1}{K} \sum_{i \in N_0} \mathbf{1}(y_i = j).$$

The observation x_0 is assigned to the class with the largest estimated probability. The output is a class label, optionally accompanied by estimated class probabilities. The natural loss function is misclassification error.

KNN regression

For a quantitative response, the prediction is the average of the responses of the K nearest neighbors:

$$\hat{f}(x_0) = \frac{1}{K} \sum_{i \in N_0} y_i.$$

This provides a local estimate of $E(Y \mid X = x_0)$. The output is a numerical value and the usual loss is squared error.

Common characteristics and differences

Both methods are nonparametric and depend on a distance metric and the value of K . A small K produces a flexible fit with lower bias but higher variance, whereas a large K produces a smoother fit with higher bias and lower variance. Both methods can suffer from the curse of dimensionality.

The key differences are therefore: (i) response type, class versus number; (ii) neighborhood summary, vote/proportion versus mean; (iii) output, label versus real-valued forecast; and (iv) usual loss, misclassification versus squared error.

2.2 Exercise 7: R^2 Equals the Square of Correlation in Simple Linear Regression

Consider simple linear regression of Y on X , and assume $\bar{x} = \bar{y} = 0$. The least-squares slope is

$$\hat{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2}, \quad \hat{y}_i = \hat{\beta}_1 x_i.$$

Since $\bar{y} = 0$, the total sum of squares is

$$TSS = \sum_i y_i^2.$$

The residual sum of squares is

$$RSS = \sum_i (y_i - \hat{\beta}_1 x_i)^2.$$

Expanding and substituting the expression for $\hat{\beta}_1$ gives

$$RSS = \sum_i y_i^2 - \frac{(\sum_i x_i y_i)^2}{\sum_i x_i^2}.$$

Therefore,

$$R^2 = 1 - \frac{RSS}{TSS} = \frac{(\sum_i x_i y_i)^2}{(\sum_i x_i^2)(\sum_i y_i^2)}.$$

Under the same centering assumption, the sample correlation is

$$\text{Cor}(X, Y) = \frac{\sum_i x_i y_i}{\sqrt{(\sum_i x_i^2)(\sum_i y_i^2)}}.$$

Squaring both sides gives

$$\text{Cor}(X, Y)^2 = \frac{(\sum_i x_i y_i)^2}{(\sum_i x_i^2)(\sum_i y_i^2)} = R^2.$$

Thus, in simple linear regression with an intercept, $R^2 = \text{Cor}(X, Y)^2$. The same result follows without the centering assumption by replacing each observation with its deviation from its sample mean.

A numerical check reported in R gives $R^2 = 0.6680$ and $\text{Cor}(X, Y)^2 = 0.6680$, agreeing to machine precision.

```
1 set.seed(7)
2 x <- rnorm(200)
3 y <- 1.5 * x + rnorm(200)
```

```

4 x <- x - mean(x)
5 y <- y - mean(y)
6 summary(lm(y ~ x))$r.squared
7 cor(x, y)^2

```

2.3 Exercise 12: Simple Linear Regression without an Intercept

(a) When are the two coefficient estimates equal?

For the no-intercept model $Y = \beta X + \epsilon$, the least-squares slope is

$$\hat{\beta}_{Y|X} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

Reversing the roles of X and Y gives

$$\hat{\beta}_{X|Y} = \frac{\sum_i x_i y_i}{\sum_i y_i^2}.$$

The numerators are identical. Therefore, apart from the trivial case where the common numerator is zero, the estimates are equal if and only if

$$\sum_i x_i^2 = \sum_i y_i^2,$$

which means that the two vectors have the same Euclidean length. Examples include $y = x$, $y = -x$, or a permutation of the values of x .

(b) Example where the estimates differ

If Y is approximately twice X , the norm of Y is much larger than the norm of X . The following simulation uses $y = 2x + \text{noise}$.

```

1 set.seed(12)
2 n <- 100
3 x <- rnorm(n)
4 y <- 2 * x + rnorm(n, sd = 0.8)
5 coef(lm(y ~ x + 0))
6 coef(lm(x ~ y + 0))

```

The fitted slopes are approximately $\hat{\beta}_{Y|X} = 2.0144$ and $\hat{\beta}_{X|Y} = 0.4103$. They are not equal. The first is close to 2, as expected from the simulation design, while the reverse regression is much smaller because $\sum y_i^2$ is substantially larger than $\sum x_i^2$.

(c) Example where the estimates are the same

Keep the same x values and let y be a random permutation of x . The two vectors then contain exactly the same numbers, so

$$\sum_i x_i^2 = \sum_i y_i^2.$$

Consequently, the two no-intercept slopes must be equal.

```

1 set.seed(12)
2 x <- rnorm(100)
3 y <- sample(x)
4 sum(x^2) == sum(y^2) # TRUE
5 coef(lm(y ~ x + 0))
6 coef(lm(x ~ y + 0))

```

Here $\sum x_i^2 = \sum y_i^2 = 74.19$, and both coefficient estimates are approximately -0.043949 . They agree, although they are not equal to 1 because a permutation of x is not necessarily the same vector as x itself.

Table 8: No-intercept slopes from the two simulated examples

Example	$\sum x_i^2 = \sum y_i^2?$	$\hat{\beta}_{Y X}$	$\hat{\beta}_{X Y}$
$y = 2x + \text{noise}$	No	2.0144	0.4103
$y = \text{permutation}(x)$	Yes	-0.0439	-0.0439

3 Conclusion

The exercises demonstrate the distinction between classification and regression, and between inference and prediction. Analysis of the **Auto** data shows strong relationships between fuel efficiency and vehicle size, weight, engine displacement, horsepower, cylinders, model year, and origin. The regression exercises further illustrate fundamental properties of least-squares estimation, including the relationship between R^2 and correlation and the behavior of no-intercept regression when the roles of the response and predictor are reversed.

References

James, G., Witten, D., Hastie, T., & Tibshirani, R. (2023). *An Introduction to Statistical Learning with Applications in R* (2nd ed., corrected June 2023). Springer.

End of work